MINI PROJECT
DOCUMENTATION

Smart Electricity Usage Analyzer

Understand energy consumption. Estimate electricity costs.
Make smarter energy-saving decisions.

Built using: Base44 AI Website Builder
Live Project:
https://refined-volt-view-pro.base44.app

PROJECT OVERVIEW

This mini project is a web-based electricity usage analysis concept designed to help users estimate energy consumption from appliance details, understand approximate electricity costs, and identify practical ways to reduce unnecessary energy use. The website was created with the assistance of Base44 AI.

Note: Calculations are estimates based on user-provided appliance power ratings and usage hours. Actual electricity-meter readings require a compatible meter or hardware integration.

1. Abstract

Electricity is essential in homes, colleges, offices, and businesses. However, many people do not know which appliances use the most electricity or how everyday usage affects their electricity bill. Smart Electricity Usage Analyzer aims to make this information easier to understand through a simple web interface. Users can enter appliance details such as wattage, quantity, and hours of use. The system can then calculate estimated energy consumption in kilowatt-hours (kWh), estimate costs using a per-unit tariff, and present the results in a clear format. The project also promotes energy-saving habits through practical recommendations.

2. Problem Statement

Many electricity consumers receive a bill without having a clear, appliance-wise understanding of their energy use. This makes it difficult to identify high-consuming devices, estimate the impact of long usage hours, and decide where energy-saving changes may be useful. A simple and accessible analysis tool can help users make more informed decisions.

3. Objectives

- Estimate electricity consumption from appliance power ratings and usage patterns.
- Calculate an approximate electricity bill using a user-entered cost per unit.
- Compare the estimated energy use of different appliances.
- Highlight appliances that contribute most to the estimated consumption.
- Provide practical suggestions for reducing unnecessary electricity use.
- Present information through a clear, responsive, and easy-to-use website.

4. Proposed Solution

The website provides a central place to enter appliance details and review calculated results. Rather than relying on confusing technical information, it converts appliance usage into familiar outputs such as units consumed and estimated cost. Visual summaries and energy-saving tips can help users understand their usage patterns and explore changes that may reduce consumption.

5. Main Features

Feature	Description
Homepage	Introduces the project, its purpose, and the main actions users can take.
Appliance input	Collects appliance name, quantity, wattage, and usage hours.
Energy calculator	Calculates estimated energy use in kWh from entered values.
Bill estimator	Multiplies estimated kWh by the entered tariff per unit.
Usage analysis	Supports appliance comparison and identification of higher estimated consumption.
Energy-saving tips	Provides practical advice for more efficient electricity use.
Visual dashboard	Can present totals and trends using summary cards or charts where implemented.

6. Technology Used

Technology / Tool	Purpose
Base44 AI Website Builder	Used to create and refine the web application through AI-assisted prompts.
Web interface	Provides the pages, forms, navigation, and results users interact with.
Calculation logic	Applies energy and estimated-cost formulas to user-entered values.
Charts / visual summaries	Used where implemented to make energy information easier to interpret.

7. Working Methodology

1. **Enter appliance information:** The user provides the appliance name, number of units, power rating in watts, hours used per day, and number of days.
2. **Calculate energy use:** The system converts the total watt-hours into kilowatt-hours (kWh).
3. **Estimate cost:** The estimated consumption is multiplied by the user's electricity tariff per unit.
4. **Compare appliances:** The system can sort or display appliance estimates to help identify larger contributors.
5. **Review recommendations:** Users review energy-saving suggestions and consider changing usage habits.

8. Important Formulas

Energy consumption:

Energy (kWh) = (Power (W) × Quantity × Hours per day × Number of days) ÷ 1000

Estimated electricity cost:

Estimated cost (■) = Energy consumption (kWh) × Cost per unit (■/kWh)

Example calculation

Suppose two 60 W fans are used for 5 hours per day for 30 days:
Energy = (60 × 2 × 5 × 30) ÷ 1000 = **18 kWh**.
If the assumed tariff is ■7 per unit, estimated energy cost = 18 × 7 = **■126**.

This is an illustrative example only. The actual bill may differ because of tariff slabs, fixed charges, taxes, meter readings, and other fees.

9. System Workflow

Step	User / System Action	Output
1	User enters appliance details.	Validated input values.
2	System applies the energy formula.	Estimated kWh per appliance.
3	System applies the tariff.	Estimated electricity cost.
4	System compares the estimates.	Usage summary and high-consumption items.
5	User reviews energy-saving guidance.	Ideas for more efficient usage.

10. Testing Checklist

- Check that valid appliance details produce the expected kWh calculation.
- Check that changing quantity, wattage, hours, or days updates the estimate correctly.
- Check that a zero or negative value is handled appropriately and invalid entries are rejected.
- Check that estimated cost changes when the tariff changes.
- Check that appliance comparison uses the calculated values.
- Check all navigation links and action buttons on desktop and mobile screens.
- Check whether entered data persists after refreshing or reopening the site, if persistence is implemented.
- Check that sample or demonstration data is clearly labelled.

11. Limitations

- Results depend on the accuracy of the wattage and usage details entered by the user.
- The website does not automatically measure real-time household electricity use unless compatible meter hardware or an external data integration is added.
- A simple per-unit estimate may not match the final utility bill when slab tariffs, fixed charges, taxes, or other adjustments apply.
- Any charts, saved history, export functions, or user accounts should be tested in the deployed application before being described as fully operational.

12. Future Enhancements

- Integrate with a smart meter or IoT energy-monitoring device for real readings.
- Add user accounts and secure storage of usage history.
- Add configurable local electricity tariffs and slab-based bill calculation.
- Provide monthly reports and downloadable summaries.
- Add alerts when estimated consumption exceeds a user-defined target.
- Improve recommendations using historical data and user-selected energy goals.

13. Conclusion

Smart Electricity Usage Analyzer demonstrates how a simple web application can make electricity consumption easier to understand. By entering appliance details, users can estimate units consumed, calculate approximate costs, compare appliances, and learn practical energy-saving habits. Built with the assistance of Base44 AI, the project provides a foundation that can be extended with more advanced analytics, tariff configuration, and real smart-meter integration.

14. Live Project

https://refined-volt-view-pro.base44.app

15. Suggested GitHub Repository Description

A Base44 AI-assisted mini project that estimates household electricity consumption and electricity costs from appliance usage details, helping users understand energy use and explore energy-saving ideas.

Suggested repository files

- **README.md** — Project overview, features, formulas, setup/use instructions, and live demo link.
- **docs/Smart_Electricity_Usage_Analyzer_Documentation.pdf** — This project report.
- **screenshots/** — Add screenshots of the homepage, analyzer, and dashboard.
- **LICENSE** — Add a license if you want to specify how others may reuse the project.

Suggested README introduction

Smart Electricity Usage Analyzer
A web-based mini project built with Base44 AI to estimate electricity consumption and costs from appliance usage details. The goal is to help users compare appliance energy use and make more informed energy-saving decisions.

Live Demo: https://refined-volt-view-pro.base44.app

Core calculation: Energy (kWh) = Power (W) × Quantity × Hours per day × Days ÷ 1000.

— End of Documentation —